incidence, $i \approx 90°$, $r = 41.8°$. From （8）, （9）,

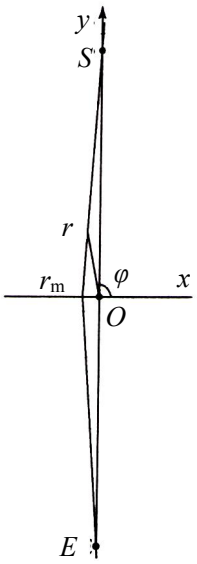
$$\varphi = 83.6° \ , \ \varphi' = 0°. \qquad (11)$$

6.

The equation

$$r = \frac{GM/c^2}{a\cos\varphi + a^2(1 + \sin^2\varphi)} \qquad (1)$$

is even in φ; the ray is symmetric about the polar axis. Left of the origin, $a < 0$; on the right, $a > 0$. The figure shows $a < 0$, with axis Ox and dwarf at O. In （1）, insert the periapsis coordinates $r = r_m$, $\varphi = \pi$ and note $a^2 \gg |a|$, giving



$$a \approx -GM/c^2 r_m \ . \qquad (2)$$

Rays past both sides of the dwarf subtend θ_S at the observer; this can be expressed in different ways, yielding different solutions.

Method 1: let d be the dwarf－Earth distance. Approximately,

$$\theta_S \approx 2r_m/d \ . \qquad (3)$$

In （1）, substitute observer coordinates $r = d$, $\varphi = -\pi/2$:

$$a^2 \approx GM/2c^2 d \ . \qquad (4)$$

Eliminate a from （2） and （4） to find

$$r_m = \sqrt{2GMd/c^2} \ . \qquad (5)$$

Substitute （5） into （3）:

$$\theta_S \approx \sqrt{8GM/c^2 d} \ ; \qquad (6)$$

Thus $$M \approx \theta_S^2 c^2 d/8G \ , \qquad (7)$$

Here $d = 3.187 \times 10^{17}$ m. Numerical substitution gives

$$M \approx 2.07 \times 10^{30} \text{kg} \ . \qquad (8)$$

Method 2: coordinates of the ray going to infinity can be written

$$r \to \infty \ , \ \varphi = -\frac{\pi}{2} + \frac{\theta}{2} \ . \qquad (9)$$

Take $\theta_S \approx \theta$, insert （9） into （1）,(set the （1） denominator to zero; note $\theta \ll 1$:

$$a\theta/2 + 2a^2 = 0 \ .$$

Thus $$\theta_S \approx \theta = -4a = \sqrt{8GM/c^2 d} \ , \qquad (10)$$

Use （4） and note $a < 0$. Equations （10）(and （6） agree, again giving （8）.